

Karl Schwede

CONTACT INFORMATION	Department of Mathematics The University of Utah 155 S 1400 E Room 233 Salt Lake City, UT 84112-0090	office: JWB 315 email: schwede@math.utah.edu web: kschwede.github.io
RESEARCH INTERESTS	I do basic research in mathematics, studying algebra, geometry and particularly singularities. Much of my work is in the setting of modular arithmetic (also known as clock arithmetic), the same setting as much of our modern communication systems. Within mathematics, I work on the boundary of the fields algebraic geometry and commutative algebra. I am also interested in number theory and computational algebra.	
EDUCATION	University of Washington , Seattle, WA <i>Ph.D., Mathematics</i> Advisor: Sándor Kovács Dissertation: On F -injective and Du Bois singularities	August 2006
	Whitman College , Walla Walla, WA <i>B.A., Mathematics (Honors), Minor: Computer Science</i>	June 1999
	Bellevue Community College , Bellevue, WA <i>Associate in Arts and Sciences</i>	June 1997
PROFESSIONAL EXPERIENCE	The University of Utah , Salt Lake City, UT <i>Professor</i>	Jul. 2018–
	<i>Associate Professor</i>	Jul. 2014–2018
	The Pennsylvania State University , State College, PA <i>Assistant Professor</i>	Jan. 2011 –Jun. 2014
	MSRI/SLMath , Berkeley, CA <i>Research Member</i>	Jan.–Mar. 2009
	<i>Research Member</i>	Apr.–May 2013
	<i>Research Member</i>	Apr.–May 2019
	<i>Research Member</i>	Apr.–May 2024
	University of Utah , Salt Lake City, UT <i>Visiting Assistant Professor</i>	Fall 2010
	Johannes Gutenberg University Mainz , Mainz Germany <i>Honorary Visiting Assistant Professor</i>	Jun. 2010
	University of Michigan , Ann Arbor, MI <i>NSF Postdoc / Postdoctoral Assistant Professor</i>	Sep. 2006 –Jul. 2010
	University of Washington , Seattle, WA <i>Teaching Assistant and Instructor</i>	Sep. 2000 –Aug. 2006
	Havas Interactive (division of Vivendi) , Walla Walla, WA <i>Computer Programmer</i> developing educational software	May 1999 –Aug. 2000

AWARDS &
FUNDING

PI – NSF, <i>Collaborative Research: Bridging Singularities in Algebra and Geometry Across Characteristics</i> . DMS-2501903	Aug 2025 – Jul 2027
PI – Simons Foundation, Travel Support for Mathematicians <i>Singularities in zero, positive, and mixed characteristic</i> . SFI-MPS-TSM-00013051	Sep 2025 – Aug 2030
University of Utah College of Science Award for Excellence in Teaching and Mentoring	April 2026
PI – NSF, <i>A Unified Perspective on Singularities in Commutative Algebra and Algebraic Geometry</i> . DMS-2101800	Sept 2021 – Aug 2025
American Mathematical Society, <i>Fellow of the AMS</i> .	Nov. 2020 – present
Simons Foundation, <i>Simons Fellow in Mathematics</i> .	Sep. 2020 – May 2021
co-PI – NSF, <i>FRG: Collaborative Research: Algebraic Geometry and Singularities in Positive and Mixed Characteristic</i> , DMS-1952522.	Jun. 2020 – May 2024
PI – NSF, <i>RTG: Algebra, Geometry, and Topology at the University of Utah</i> , DMS-1840190.	June 2019 – May 2025
PI – NSF, <i>Commutative Algebra: Singularities in All Characteristics with Geometric Applications</i> , DMS-1801849.	Sep. 2018 – Aug. 2021
University of Utah, <i>Presidential Scholar</i> .	Sep. 2017 – Aug. 2020
PI – NSF, <i>CAREER: Test Ideals and the Geometry of Projective Varieties in Positive Characteristic</i> , DMS-1252860/1501102.	Sep. 2013 – Aug. 2019
PI – NSF, <i>FRG: Collaborative Research: Birational Geometry and Singularities in Zero and Positive Characteristic</i> , DMS-1265261/1501115.	July 2013 – June 2017
co-PI – NSF, for conference <i>Macaulay2 Development workshop</i> , DMS- 1601205.	May 2016 – April 2017
Alfred P. Sloan Foundation, <i>Alfred P. Sloan Research Fellowship</i> .	Sept. 2012 – Sept. 2016
co-PI – NSF, for conference <i>Computational Workshop on Singularities and Invariants Defined by Frobenius</i> , DMS-1160927.	May 2012
PI – NSF, <i>Singularities in Characteristic Zero and Singularities in Positive Characteristic</i> , DMS-0969145/1064485.	Sept. 2010 – Aug. 2013
PI then co-PI – NSF, for conference <i>Frobenius splitting in algebraic geometry, commutative algebra, and representation theory</i> , DMS-0968646.	May 2010
PI – NSF, Postdoctoral Fellowship, DMS-0703505.	Sept. 2007 – Aug. 2010

1. *Gluing Schemes and a Scheme Without Closed Points*. Recent progress in arithmetic and algebraic geometry, 157–172, Contemp. Math. 386. 2005
2. *A simple characterization of Du Bois singularities*. Compos. Math. 143, no. 4, 813–828. 2007
3. *Rational singularities associated to pairs*, with S. Takagi. Michigan Math. J. 57, 625–658. 2008
4. *Generalized test ideals, sharp F -purity, and sharp test elements*. Math. Res. Lett. 15, no. 6, 1251–1261. 2008
5. *F -injective singularities are Du Bois*. Amer. J. Math. 131, no. 2, 445–473. 2009
6. *F -adjunction*. Algebra & Number Theory. 3, no. 8, 907–950. 2009
7. *The canonical sheaf of Du Bois singularities*, with S. Kovács and K. Smith. Adv. Math. 224, no. 4, 1618–1640. 2010
8. *Globally F -regular and log Fano varieties*, with K. Smith. Adv. Math. 224, no. 3, 863–894. 2010
9. *A refinement of sharply F -pure and strongly F -regular pairs*. Journal of Commutative Algebra, 2, no. 1, 91–110, 2010
10. *Centers of F -purity*. Math. Z. 265, no. 3, 687–714. 2010
11. *Discreteness and rationality of F -jumping numbers on rings with singularities*, with M. Blickle, S. Takagi and W. Zhang. Math. Ann., 347, no. 4, 917–949. 2010.
12. *On the number of compatibly Frobenius split subvarieties, prime F -ideals, and log canonical centers*, with K. Tucker. Ann. Inst. Fourier (Grenoble) **60** (2010), no. 5, 1515–1531.
13. *Hodge theory meets the minimal model program: a survey of log canonical and Du Bois singularities*, with S. Kovács. Topology of Stratified Spaces (G. Friedman, E. Hunsicker, A. Libgober, and L. Maxim, eds.), Math. Sci. Res. Inst. Publ., vol. 58, Cambridge Univ. Press, Cambridge, 2011, pp. 51–94.
14. *Test ideals in non- $\mathbb{Q}$ -Gorenstein rings*, Trans. Amer. Math. Soc. **363** (2011), no. 11, 5925–5941
15. *A note on discreteness of F -jumping numbers*. Proc. Amer. Math. Soc. **139** (2011), no. 11, 3895–3901
16. *Supplements to non-LC ideal sheaves*, with O. Fujino and S. Takagi. Higher Dimensional Algebraic Geometry, RIMS Kôkyûroku Bessatsu, B24, Res. Inst. Math. Sci. (RIMS), Kyoto, 2011, pp. 1–47.
17. *Semi-log canonical vs F -pure singularities*, with L. E. Miller. J. Alg. **349**, (2012), no. 1, 150–164.
18. *On the behavior of test ideals under finite morphisms*, with K. Tucker. J. Algebraic Geom. 23 (2014), no. 3, 399–443.
19. *Test ideals via a single alteration and discreteness and rationality of F -jumping numbers*, with K. Tucker and W. Zhang. Math. Res. Lett. 19, (2012), no. 01, 191–197.
20. *A survey of test ideals*, with K. Tucker. Progress in Commutative Algebra 2, Closures, Finiteness and Factorization, Walter de Gruyter GmbH & Co. KG, Berlin, (2012), 39–99.

21. *An algorithm for computing compatibly Frobenius split subvarieties*, with M. Katzman. J. Symbolic Comput., **47**, (2012), no. 8, 996-1008.
22. *Cartier modules on toric varieties*, with J.-C. Hsiao, and W. Zhang. Trans. Amer. Math. Soc. 366 (2014), no. 4, 1773-1795.
23. *Du Bois singularities deform*, with S. Kovács. in *Minimal Models and Extremal Rays (Kyoto, 2011)*, Adv. Stud. Pure Math. (2016), 70, 49–66.
24. *A canonical linear system associated to adjoint divisors in characteristic $p > 0$* . J. Reine Angew. Math. 696 (2014), 69-87.
25. *p^{-1} -linear maps in algebra and geometry*, with M. Blickle. Commutative Algebra, Expository Papers Dedicated to David Eisenbud on the Occasion of His 65th Birthday (I. Peeva ed.), Springer New York Heidelberg Dordrecht London, 2013, pp. 123–205.
26. *F-signature of pairs and the asymptotic behavior of Frobenius splittings*, with M. Blickle and K. Tucker. Adv. Math. **231**, (2012) no. 6, 3232-3258.
27. *F-signature of pairs: Continuity, p -fractals and minimal log discrepancies*, with M. Blickle and K. Tucker. J. London Math. Soc. **87** (2013), no. 3, 802–818.
28. *Richardson varieties have Kawamata log terminal singularities*, with S. Kumar. Int. Math. Res. Not. IMRN 2014, no. 3, 842-864.
29. *Bertini theorems for F-singularities*, with W. Zhang. Proc. Lond. Math. Soc. (3) 107 (2013), no. 4, 851-874.
30. *A dual to tight closure theory*, with N. Epstein. Nagoya Math. J. 213 (2014), 41-75.
31. *Depth of F-singularities and base change of relative canonical sheaves*, with Z. Patakfalvi. Journal of the Institute of Mathematics of Jussieu. **13**, no. 1, (2014) 43–63.
32. *On the numerical dimension of pseudo-effective divisors in positive characteristic*, with P. Cascini, C. Hacon and M. Mustața. Amer. J. Math. 136 (2014), no. 6, 1609-1628.
33. *Appendix: F-injectivity and depth*, with A. K. Singh. Appendix to *Deformations of F-injectivity and local cohomology* by J. Horiuchi, L. E. Miller and K. Shimomoto. Indiana Univ. Math. J. 63 (2014), no. 4, 1139-1157.
34. *A Frobenius variant of Seshadri constants*, with M. Mustața. Math. Ann. 358 (2014), no. 3-4, 861-878.
35. *Explicitly extending Frobenius splittings over finite maps*, with K. Tucker. Comm. Algebra 43 (2015), no. 10, 4070-4079.
36. *Rings of Frobenius operators*, with M. Katzman, A. K. Singh and W. Zhang. Math. Proc. Cambridge Philos. Soc. 157 (2014), no. 1, 151-167.
37. *Test ideals of non-principal ideals: Computations, Jumping Numbers, Alterations and Division Theorems*, with K. Tucker. J. Math. Pures Appl. (9) 102 (2014), no. 5, 891-929.
38. *F-singularities via alterations*, with M. Blickle and K. Tucker. Amer. J. Math. 137 (2015), no. 1, 61–109.
39. *The weak ordinarity conjecture and F-singularities*, with B. Bhatt and S. Takagi. Adv. Stud. Pure Math., 74, Math. Soc. Japan, Tokyo, 2017.

40. *Uniform bounds for strongly F -regular surfaces*, with Paolo Cascini and Yoshinori Gongyo. Trans. Amer. Math. Soc. 368 (2016), no. 8, 5547–5563.
41. *On rational connectedness of globally F -regular threefolds*, with Y. Gongyo, Z. Li, Z. Patakfalvi, H. Tanaka, and R. Zong. Adv. Math. 280 (2015), 47–78.
42. *Inversion of adjunction for rational and Du Bois pairs*, with S. Kovács. Algebra & Number Theory. 10 (2016), no. 5, 969–1000.
43. *Test ideals in rings with finitely generated anti-canonical algebras*, with A. Chiechio, F. Enescu and L. E. Miller. J. Inst. Math. Jussieu 17 (2018), no. 1, 171–206.
44. *The F -different and a canonical bundle formula*, with O. Das. Ann. Sc. Norm. Super. Pisa Cl. Sci. 17 (2017), no 3, 1173–1205
45. *F -singularities in families*, with Z. Patakfalvi and W. Zhang. Algebr. Geom. 5 (2018), no. 3, 264–327.
46. *Positive characteristic algebraic geometry*, with Z. Patakfalvi and K. Tucker. Surveys on recent developments in algebraic geometry, 33–80, Proc. Sympos. Pure Math., 95, Amer. Math. Soc., Providence, RI, 2017.
47. *On the behavior of singularities at the F -pure threshold*, with E. Canton, D. Hernández and E. Witt. Appendix by Alessandro De Stefani, Jack Jeffries, Zhibek Kadyrsizova, Robert Walker, George Whelan. Illinois J. Math. 60 (2016), no. 3–4, 669–685
48. *The dualizing complex of F -injective and Du Bois singularities*, with B. Bhatt and L. Ma. Math. Z. 288 (2018), no. 3–4, 1143–1155.
49. *Discreteness of F -jumping numbers at isolated non- Q -Gorenstein points*, with P. Graf. Proc. Amer. Math. Soc. 146 (2018), no. 2, 473–487.
50. *Local cohomology of Du Bois singularities and applications to families.*, with L. Ma and K. Shimomoto. Compos. Math. 153 (2017), no. 10, 2147–2170.
51. *Fundamental groups of F -regular singularities via F -signature*, with J. Carvajal-Rojas and K. Tucker. Ann. Sci. Éc. Norm. Supér. (4) 51 (2018), no. 4, 993–1016.
52. *Étale fundamental groups of strongly F -regular schemes*, with B. Bhatt, J. Carvajal-Rojas, P. Graf and K. Tucker. Int. Math. Res. Not. IMRN 2019, no. 14, 4325–4339.
53. *Divisor package for Macaulay2*, with Z. Yang. J. Softw. Algebra Geom. 8 (2018), 87–94.
54. *Perfectoid multiplier/test ideals in regular rings and bounds on symbolic powers*, with L. Ma. Invent. Math. 214 (2018), no. 2, 913–955.
55. *F -signature under birational morphisms*, with L. Ma., T. Polstra, K. Tucker. Forum Math. Sigma 7 (2019), e11.
56. *The TestIdeals package for Macaulay2*, with A. F. Boix, D. J. Hernández, Z. Kadyrsizova, M. Katzman, S. Malec, M. Robinson, D. Smolkin, P. Teixeira, E. E. Witt. J. Softw. Algebra Geom. 9 (2019), no. 2, 89–110.
57. *Recent applications of p -adic methods to commutative algebra*, with L. Ma. Notices Amer. Math. Soc. 66 (2019), no. 6, 820–831.
58. *Seminormalization package for Macaulay2*, with B. Serbinowski. J. Softw. Algebra Geom. 10 (2020), no. 1, 1–7.

PUBLICATIONS
CONTINUED
ALL HAVE BEEN
REFEREED

59. *A Kunz-type characterization of regular rings via alterations*, with L. Ma. J. Pure Appl. Algebra 224 (2020), no. 3, 1124–1131.
60. *Singularities in mixed characteristic via Perfectoid big Cohen-Macaulay algebras*, with L. Ma. Duke Math. J. 170 (2021), no. 13, 2815–2890.
61. *The FrobeniusThresholds package for Macaulay2*, with D. J. Hernández, P. Teixeira, E. E. Witt. J. Softw. Algebra Geom. 11 (2021), no. 1, 25–39.
62. *Bertini Theorems for F -signature*, with J. Carvajal-Rojas and K. Tucker. Math. Z. 299 (2021), no. 1-2, 1131–1153.
63. *Covers of rational double points in mixed characteristic*, with J. Carvajal-Rojas, L. Ma, T. Polstra, K. Tucker. Covers of rational double points in mixed characteristic. J. Singul. 23 (2021), 127–150.
64. *Maximal Cohen-Macaulay complexes and their uses: A partial survey*, with S. Iyengar, L. Ma, M. Walker. arXiv:2106.08173. Commutative algebra, 475–500, Springer, Cham, (2021).
65. *An analog of adjoint ideals and PLT singularities in mixed characteristic*, with L. Ma, K. Tucker, J. Waldron, J. Witaszek. J. Algebraic Geom. 31 (2022), no. 3, 399–443.
66. *RationalMaps, a package for Macaulay2*, with C.J. Bott, S. H. Hassanzadeh, D. Smolkin. J. Softw. Algebra Geom. 12 (2022), no. 1, 17–26.
67. *Symbolic power containments in singular rings in positive characteristic*, with E. Grifo and L. Ma. Manuscripta Math. 170 (2023), no. 3-4, 471–496.
68. *Compatible ideals in Gorenstein rings*, with T. Polstra. Proc. Amer. Math. Soc. 151 (2023), no. 10, 4099–4112.
69. *Finding points on varieties with Macaulay2*, with S. Bisui, S. Maitra, T. T. Nguyễn. J. Softw. Algebra Geom. 13 (2023), no. 1, 33–43.
70. *FastMinors package for Macaulay2*, with B. Martinova, M. Robinson, Y. Yao. arXiv:2002.05758. J. Softw. Algebra Geom. 13 (2023), no. 1, 13–31.
71. *Globally $+$ -regular varieties and the minimal model program for threefolds in mixed characteristic*, with B. Bhatt, L. Ma, Z. Patakfalvi, K. Tucker, J. Waldron, J. Witaszek. Publ. Math. Inst. Hautes Études Sci. 138 (2023), 69–227.
72. *Global generation of test ideals in mixed characteristic and applications*, with C. Hacon, A. Lamarche. Algebr. Geom. 11 (2024), no. 5, 676–711.
73. *Finite generation of split F -regular monoid algebras*, with R. Datta, K. Tucker. J. Lond. Math. Soc. (2) 112 (2025), no. 1, Paper No. e70234, 55 pp.

ACCEPTED
PAPERS

74. *The Briançon-Skoda Theorem via weak functoriality of big Cohen-Macaulay algebras*, with S. Rodríguez-Villalobos. To appear in Michigan Math. J., arXiv:2406.02433.
75. *Plus-pure thresholds of some cusp-like singularities in mixed characteristic*, with H. Cai, S. Pande, E. Quinlan-Gallego, K. Tucker. To appear in J. Comm. Algebra, arXiv:2501.07528.
76. *Perfectoid signature, perfectoid Hilbert-Kunz multiplicity, and an application to local fundamental groups*, with H. Cai, S. Lee, L. Ma, K. Tucker. To appear in Invent. Math. arXiv:2209.04046, <https://doi.org/10.1007/s00222-025-01399-w>.

ACCEPTED
PAPERS
CONTINUED

77. *Bounds on the plus-pure thresholds of some hypersurfaces in (ramified) regular rings*, M. Benozzo, V. Jagathese, V. Pandey, P. Ramírez-Moreno, P. Sridhar. To appear in Math. Z. [arXiv:2509.07217](#).

SUBMITTED
PAPERS
/PREPRINTS

78. *Test ideals in mixed characteristic: a unified theory up to perturbation*, with B. Bhatt, L. Ma, Z. Patakfalvi, K. Tucker, J. Waldron, J. Witaszek. [arXiv:2401.00615](#).

79. *Perfectoid pure singularities*, with B. Bhatt, L. Ma, Z. Patakfalvi, K. Tucker, J. Waldron, J. Witaszek. [arXiv:2409.17965](#).

80. *BCM-thresholds of non-principal ideals*, with S. Rodríguez-Villalobos. [arXiv:2501.16773](#).

81. *Closure operations induced via resolutions of singularities in characteristic zero*, with N. Epstein, P. M. McDonald, R. R.G. [arXiv:2504.05554](#).

82. *Variants on Frobenius Intersection Flatness and Applications to Tate Algebras*, with N. Epstein, R. Datta, K. Tucker. [arXiv:2504.06444](#).

83. *The Briançon-Skoda theorem for pseudo-rational and Du Bois singularities and uniformity in excellent rings*, with L. Ma, P. McDonald, R. R.G. [arXiv:2510.11540](#).

84. *F-finite schemes have a dualizing complex*, with Bhargav Bhatt, Manuel Blickle, Kevin Tucker. [arXiv:2604.20005](#).

BOOK DRAFT

85. *Singularities defined by Frobenius and applications*, with K. E. Smith. Book draft. The latest version is available here: <https://github.com/kschwede/FrobeniusSingularitiesBook>.

SOFTWARE

RationalMaps package for Macaulay2, a package for checking whether a rational map is birational/regular/an embedding. With C.J. Bott, H. Hassanzadeh, and D. Smolkin. The current version is in the Macaulay2 build tree: <https://github.com/Macaulay2/M2/tree/master/M2/Macaulay2/packages> **2016–present**

Pullback package for Macaulay2, a package for computing pullbacks in the category of rings. With D. Ellingson. It is part of the current Macaulay2 build tree: <https://github.com/Macaulay2/M2/tree/master/M2/Macaulay2/packages> **2015–present**

Seminormalization package for Macaulay2, a package for computing seminormalizations. With B. Serbinowski. The current version is in the Macaulay2 build tree: <https://github.com/Macaulay2/M2/tree/master/M2/Macaulay2/packages> **2018–present**

Divisor package for Macaulay2, a package for computations with Weil divisors on normal varieties. With Z. Yang. It is part of the current Macaulay2 build tree: <https://github.com/Macaulay2/M2/tree/master/M2/Macaulay2/packages> **2014–present**

Software continued on the next page.

SOFTWARE CONTINUED	<i>TestIdeals package</i> for Macaulay2, a package for computing F-singularities (test ideals, F-rationality, etc.). With E. Bela, A. F. Boix, J. Bruce, D. Hernandez, Z. Kadyrsizova, M. Katzman, S. Malec, M. Robinson, D. Smolkin, P. Teixeira and E. Witt. The latest stable version of <code>TestIdeals</code> is in the Macaulay2 build tree: https://github.com/Macaulay2/M2	2012–present
	Macaulay2 function for computing compatibly split subvarieties, With Mordechai Katzman. Download the current version: http://www.math.utah.edu/~schwede/M2/FSplitting.m2	2012–present
	<i>FrobeniusThresholds package</i> for Macaulay2, a package for computing F-thresholds and related invariants. With J. Bruce, D. Hernandez, D. Smolkin, P. Teixeira and E. Witt. The latest stable version of <code>FrobeniusThresholds</code> is in the Macaulay2 build tree: https://github.com/Macaulay2/M2 The bleeding edge version and related packages are in: https://github.com/kschwede/Workshop-2017-Berkeley/tree/master/Fsing/	2018–present
	<i>FastMinors package</i> for Macaulay2, faster function field linear algebra (with applications to singularities). With B. Martinova, M. Robinson, Y. Yao. The stable version is in the Macaulay2 build tree: https://github.com/Macaulay2/M2 The bleeding edge version is available at: https://github.com/kschwede/M2/blob/master/M2/Macaulay2/packages/	2019–present
	<i>RandomPoints package</i> for Macaulay2, a package for finding rational points on varieties over finite fields. With Sankhaneel Bisui, Sarasij Maitra, Thai Nguyen, Zhan Jiang. The latest stable version of <code>RandomPoints</code> is in the Macaulay2 build tree: https://github.com/Macaulay2/M2	2020–present
	<i>NonPrincipalTestIdeals package</i> for Macaulay2, a package for computing test ideals of non-principal ideals. With Trung Chau and Hunter Simper, and the contributions of Rahul Ajit and Matthew Bertucci. The latest stable version of <code>NonPrincipalTestIdeals</code> is in the Macaulay2 build tree: https://github.com/Macaulay2/M2	2024–present

CONFERENCE ORGANIZATION	Organizer for the conference <i>Notions of Singularity in Different Characteristics</i> , Banff Canada (BIRS)	Oct. 2025
	Organizer for the conference <i>Singularities in Algebra and Geometry</i> , CIMAT, Guanajuato City, Mexico	Jun. 2025
	Organizer for the Conference <i>Moduli of Varieties</i> , University of Utah	Nov. 2024
	Organizer for the Fall School <i>Methods in Mixed Characteristic Geometry</i> , Mainz Germany	Oct. 2024
	Faculty co-organizer of BRIDGES (Building Relationships for an Inclusive and Diverse Group of Emerging Students), Organized by the U. of Utah chapter of the AWM, funded by the RTG grant, DMS-1840190.	July 2024
	On the scientific committee for the conference <i>Epiga 2024</i> , Sorbonne Université, Paris	Jun. 2024
	Organizer for the conference and summer school <i>Macaulay2, Computational Algebraic Geometry and String Theory</i> , to be held at the University of Utah	May-Jun. 2024
	Lead organizer for the workshop <i>Recent Developments in Commutative Algebra</i> , to be held in MSRI During the special Commutative Algebra Semester	Apr. 2024
	On the organizing committee of <i>The Fellowship of the Ring</i> , national commutative algebra online seminar. https://sites.google.com/view/fellowship-of-the-ring	Apr. 2020 – present
	Co-organizer of an AMS Special Session, held at the University of Utah	Oct. 2022
	Co-organizer of the conference: <i>Advances in Mixed Characteristic Commutative Algebra and Geometric Connections</i> , at Oaxaca (BIRS)	May 2022
	Co-organizer of the virtual special month on Singularities and K -stability at the University of Utah, https://sites.google.com/view/special-month-on-singularities	May-Jun. 2021
	Co-organizer of an AMS Special Session, held at the University of Michigan	Oct. 2018
	Co-organizer of an AMS Special Session, held at Portland State University	Apr. 2018
	Co-organizer of an AMS Special Session, held that the Joint Mathematics Meetings in San Diego	Jan. 2018
	Co-organizer of the conference: <i>Higher dimensional algebraic geometry and characteristic $p > 0$</i> , held at CIRM / Luminy	Sept. 2016
	Co-organizer of the summer school and conference: <i>Higher Dimensional Algebraic Geometry</i> held at the University of Utah	Jul. 2016

Conference organization continued on the next page.

CONFERENCE ORGANIZATION CONTINUED	Co-organizer of the workshop: <i>Intensive Workshop for Macaulay2 Development</i> , held at the University of Utah	May 2016
	Co-organizer of the AMS special session: <i>Algebraic Geometry</i> , held at the University of Utah	Apr. 2016
	Co-organizer of the AMS-AWM special session: <i>Commutative Algebra and Its Interactions with Algebraic Geometry</i> , held at the Joint Mathematical Meetings	Jan. 2016
	Co-organizer of the <i>Mathematical Research Communities Workshop on Commutative Algebra</i> held at Snowbird Utah	Jun. 2015
	Co-organizer of the <i>Positive Characteristic Algebraic Geometry Workshop</i> held at the University of Illinois at Chicago	Mar. 2014
	Co-organizer of the AMS special session: <i>Homological and characteristic p methods in commutative algebra</i> , held at the Joint Mathematical Meetings, Baltimore	Jan. 2014
	Co-organizer of the AMS special session: <i>Special Session on The Geometry of Algebraic Varieties</i> , held at Temple University	Oct. 2013
	Co-organizer of a mini-symposium at: <i>SIAM conference on Applied Algebraic Geometry</i> , held at Colorado State University	Aug. 2013
	Co-organizer of the conference: <i>Computational workshop on Frobenius singularities and invariants</i> , held at the University of Michigan, see http://sites.google.com/site/computingfinvariantsworkshop/	May 2012
	Co-organizer of the AMS Special Session: <i>Singularities in Commutative Algebra and Algebraic Geometry</i> , held at the University of Kansas	Mar. 2012
	Co-organizer of the conference: <i>Relating test ideals and multiplier ideals</i> , held at the American Institute of Mathematics	Aug. 2011
	Co-organizer of the conference: <i>Frobenius splitting in algebraic geometry, commutative algebra, and representation theory</i> , see http://sites.google.com/site/frobeniussplitting/	May 2010
	Assistant to the organizers of <i>Commutative algebra</i> AMS MRC summer conference/school in Snowbird Utah	Jun. 2010
	Local organizer for Mel Hochster's 65th birthday conference.	Aug. 2008

SEMINAR ORGANIZATION	Organizer/co-organizer of the Utah Commutative algebra seminar.	Various years 2014–present
	Organizer/co-organizer of the Utah Algebraic Geometry seminar.	Various years 2014–present
	Organizer/co-organizer of the Penn State algebra and number theory seminar.	2013–2014
	Organizer/co-organizer of the University of Michigan commutative algebra seminar.	2008–2010
	Organizer/co-organizer of the University of Michigan topics in algebraic geometry seminar.	2007–2008
	Organizer of the University of Washington algebraic geometry seminar.	2004–2005
	Co-organizer of the University of Washington undergraduate mathematical sciences seminar	2004–2005
EDITORIAL PROFESSIONAL ACTIVITIES	Associate editor for the Journal of Singularities. https://www.journalofsing.org/	2024–present
	Editor for the journal Épijournal de Géométrie Algébrique. https://epiga.episciences.org/	2021–present
	Series Editor for the RSME Springer Series. http://www.springer.com/series/13759	2015–present
UNIVERSITY SERVICE ACTIVITIES	Mathematics Department RPT (Retention, Promotion, Tenure), Elected Chair	2023–2025
	Special Assistant to the Dean of the College of Science, University of Utah.	2020–2022
	Director of Graduate Studies, University of Utah.	2015–2017
	Member of the following committees at the University of Utah:	
	Mathematics Computing Committee.	2025–2026
	Mathematics tenured faculty review subcommittee.	2025–2026
	Pre-tenure informal review committee.	2024–2025
	Postdoc Hiring Committee.	2023–2024
	Faculty Review Committee, Mathematics Department	2022–2023
	Executive Committee, Mathematics Department	2017–2019, 2022–2024
	Senate IT Committee (SACIT).	2021–present
	Mathematics EDI committee.	2021–2022
	Hiring committee.	2021–2022, 2024–2025

University service activities continued on the next page.

UNIVERSITY SERVICE ACTIVITIES CONTINUED	College of Science EDI committee.	2021–2022
	Graduate Recruitment Committee.	2014–2017, 2019
	Graduate Committee.	2014–2017
	Outstanding Graduate Student & Instructor Award Committee.	2015–2017
	Outstanding dissertation award committee (Graduate School)	Summer 2016
	Ad hoc committee on thesis formatting (Graduate School)	2016–2019
	University TAs hip award committee (Graduate School)	Spring 2017
	Bachelor of Undergraduate Studies (BUS) Committee	2018–2020
	Departmental Thesis Standards Committee	2018–2020
	Various ad hoc committees (ie, promotion and tenure).	Various years
OTHER PROFESSIONAL ACTIVITIES	Member of the following committees at Penn State:	
	Qualifying Exam Committee, Graduate Student Teaching Committee, Library Committee, Eberly College Outreach Council.	2011–2014
	Ran a research group for the <i>Apprenticeship program in commutative algebra</i> , Fields Institute, Toronto	Jan. 2025
	Ran a 3-week long math camp for high school students, <i>Explorations in Number Theory and Cryptography</i> , University of Utah	June 2016
	Ran a week-long math camp for high school students, <i>Spy Games: the math of secret messages</i> , within Penn State’s <i>Science-U</i> camp program.	June 2013 and July 2014
	Helped build and maintain <i>Situs Geometriae Algebraicae</i> , a website designed to help students in algebraic geometry find references.	2004–2008
	(Now somewhat) active participant on http://www.mathoverflow.net .	2010–present
	Reviewed papers for AMS Math Reviews and Zentralblatt Math.	2006–present
	Helped develop the University of Washington VIGRE website.	2002–2003
	Panelist for NSF grant applications (various sorts)	Various years
	Referee for NSA grant applications.	Various years
	Referee for Simons Foundation grant applications.	Various years
	Referee for grant/fellowship applications from other institutions/foundations (ERC/DFG/etc.).	Various years
	Refereed and reviewed papers for numerous journals and conference proceedings	2007 –present

TEACHING

Instructor: I have been an instructor for:

- high school level algebra, University of Washington.
- precalculus, University of Washington.
- calculus, University of Washington, University of Michigan, University of Utah.
- multivariable calculus (Stokes theorem etc.), University of Washington.
- topology, University of Washington.
- linear algebra, University of Michigan.
- honors calculus (essentially introduction to real analysis), University of Michigan.
- introduction to topology, University of Washington.
- introduction to schemes and cohomology, University of Michigan..
- topics courses on algebraic geometry and commutative algebra, University of Utah.
- undergraduate abstract algebra, Penn State University & University of Utah.
- graduate-level commutative algebra, Penn State University & University of Utah.
- discrete mathematics (intro. proofs), Penn State University & University of Utah.
- honors multivariable calculus, University of Utah.
- graduate abstract algebra I & II, University of Utah & Penn State University.
- cryptography, University of Utah.
- introductory algebraic geometry, University of Utah.
- introduction to real analysis, University of Utah.
- undergraduate abstract algebra I & II, University of Utah.

STUDENTS & MENTORING

Ph.D. students.

- Andrew Bydlon (Summer 2017, University of Utah, transferred from PSU)
- Javier Carvajal-Rojas (Summer 2018, University of Utah)
- Daniel Smolkin (Summer 2019, University of Utah)
- Marcus Robinson (Summer 2020, University of Utah)
- Seungsu Lee (Summer 2023, University of Utah)
- Peter McDonald, joint with Srikanth Iyengar (Summer 2024, University of Utah)
- Matthew Bertucci, joint with Sean Howe (Summer 2025, University of Utah)
- Hanlin Cai, joint with Sean Howe (Summer 2024, University of Utah)
- Yotam Svoray (Spring 2026, University of Utah)
- Sandra Rodríguez-Villalobos (Summer 2026, University of Utah, expected)
- Rahul Ajit, joint with Christopher Hacon (Summer 2026, University of Utah, expected)
- Anne Fayolle (Summer 2027, University of Utah, expected)
- Gari Chua (Summer 2029, University of Utah, expected)

Masters students.

- Faith Pearson (Spring 2021, University of Utah)
- Ruyi Ma (Fall 2022, University of Utah)

Postdocs.

- Linquan Ma (2015-2018)
- Thomas Polstra (2017-2020)
- Alicia Lamarche, co-mentored with Aaron Bertram, (2020-2024)
- Eamon Quinlan, co-mentored with Anurag Singh, (2021-2025 expected)
- Quentin Posva, co-mentored with Christopher Hacon, (2022-2023)
- Suchitra Pande, (2024-2027 expected)

SELECTED
TALKS

University of Washington algebra seminar	February 2004
University of Michigan algebraic geometry seminar	November 2005
University of Washington algebra seminar	January 2006
Bellingham algebraic geometry seminar (BAGS)	February 2006
Recent Trends in Higher Dimensional Geometry, conference in Banff Canada (BIRS)	April 2006
Rice algebraic geometry seminar	March 2006
University of Michigan algebraic geometry seminar	November 2006
Davidson College AMS meeting, special session in commutative algebra	March 2007
F -singularities and D -modules conference in Ann Arbor	August 2007
University of Illinois at Chicago algebraic geometry seminar	February 2008
Purdue University working algebraic geometry seminar	February 2008
University of Georgia algebraic geometry seminar	April 2008
Conference in honor of Mel Hochster's 65th birthday	August 2008
University of Washington algebra seminar	September 2008
Vancouver British Columbia AMS Meeting, special session in algebraic geometry	October 2008
University of Illinois at Urbana-Champaign algebraic geometry seminar	October 2008
University of Utah commutative algebra seminar	November 2008
Western algebraic geometry seminar (WAGS conference)	November 2008
Syracuse University, colloquium	January 2009
University of North Carolina, colloquium	February 2009
University of Illinois at Urbana Champaign AMS meeting, special session in commutative algebra	March 2009
University of Illinois at Urbana-Champaign, colloquium	April 2009
Purdue University algebraic geometry seminar	April 2009
Commutative Algebra and its Connections to Geometry (honoring Wolmer Vasconcelos), Pan-American Advanced Study Institute, Olinda Brazil	August 2009
University of Kansas, algebra seminar	September 2009
Higher Dimensional Algebraic Geometry conference, Research Institute for Mathematical Sciences, Kyoto University, Japan	December 2009

Talks continued on the next page.

SELECTED TALKS CONTINUED	Joint mathematical meetings in San Francisco, special session on commutative algebra	January 2010
	Corcordia University, colloquium	January 2010
	Washington University, colloquium	January 2010
	Indiana University Bloomington, algebra seminar	January 2010
	Indiana University Bloomington, colloquium	January 2010
	University of Missouri, colloquium	January 2010
	Louisiana State University, colloquium	February 2010
	Wayne State University, colloquium	February 2010
	Penn State University, colloquium	February 2010
	Rice University, colloquium	February 2010
	Sheffield University, lecture series	March 2010
	Lexington Kentucky AMS Meeting, special session in commutative algebra	March 2010
	Harvard-MIT algebraic geometry seminar	April 2010
	Frobenius splitting conference	May 2010
	Algebra seminar, Johannes Gutenberg-Universität Mainz	June 2010
	Commutative Algebra in the Southeast, Conference, Atlanta Georgia	September 2010
	University of Utah, Colloquium	September 2010
	Purdue University working algebraic geometry seminar	October 2010
	Purdue University commutative algebra seminar	October 2010
	CIRM, Luminy, Commutative algebra and its interactions with algebraic geometry	November 2010
	Western algebraic geometry seminar (WAGS conference)	November 2010
	Joint mathematical meetings in New Orleans, special session on commutative algebra	January 2011
	Penn State University, algebra and number theory seminar	February 2011
	Berkeley-Davis-Stanford Algebraic Geometry Colloquium	March 2011
	University of Michigan commutative algebra seminar	March 2011
	Midwest Commutative Algebra and Geometry Conference	May 2011
	University of Washington, algebra and algebraic geometry seminar	May 2011

Talks continued on the next page.

SELECTED TALKS CONTINUED	Workshop on Almost Purity, University of Michigan	May 2011
	University of Osnabrück, college seminar - combinatorial structures in algebra and topology	June 2011
	Johannes Gutenberg-Universität Mainz - algebra seminar	July 2011
	University of Nebraska - algebraic geometry seminar	August 2011
	Route 81 Conference on Commutative Algebra and Algebraic Geometry - Cornell University	September 2011
	Stony Brook University, algebra, geometry and physics seminar	September 2011
	University of Michigan commutative algebra seminar	October 2011
	University of Utah AMS meeting, special session in commutative algebra	October 2011
	Sheffield University, lecture series	January 2012
	Universitat de Barcelona / Universitat Politècnica de Catalunya (joint seminar), Seminari de Geometria Algebraica	January 2012
	Penn State University, GAP Seminar	February 2012
	University of Illinois at Chicago, algebraic geometry seminar	March 2012
	Penn State University, algebra and number theory seminar	March 2012
	Princeton University, algebraic geometry seminar	March 2012
	Johns Hopkins University, algebraic geometry/number theory seminar	March 2012
	ACC for minimal log discrepancies and termination of flips conference, American Institute of Mathematics	May 2012
	char- p & p -adic geometry conference, Mainz Germany	June 2012
	University of Utah, algebraic geometry seminar	September 2012
	Ohio State University, algebraic geometry seminar	September 2012
	Georgia State University, colloquium	October 2012
	Georgia State University, commutative algebra seminar	October 2012
	Johns Hopkins University, algebraic geometry/number theory seminar	November 2012
	Princeton University, algebraic geometry seminar	November 2012
	Trends in Arithmetic Geometry, Lorentz Center, Leiden, the Netherlands	January 2013
	Penn State, algebra and number theory seminar	February 2013
	University of Washington, algebra and algebraic geometry seminar	March 2013

Talks continued on the next page.

SELECTED TALKS CONTINUED	Columbia University, algebraic geometry seminar	March 2013
	Imperial College of London, 6 lectures, Workshop: Characteristic p methods in algebraic geometry	April 2013
	Southern California Algebraic Geometry Seminar, a joint meeting of Caltech, UC Los Angeles, UC San Diego, and the University of Southern California	April 2013
	American Institute of Mathematics, conference on The minimal model program in characteristic p .	May 2013
	Math Science Research Institute, Conference: The Commutative Algebra of Singularities in Birational Geometry: Multiplier Ideals, Jets, Valuations, and Positive Characteristic Methods	May 2013
	CIRM Luminy, Commutative algebra and its interactions with algebraic geometry	July 2013
	Penn State University, algebra and number theory seminar	August 2013
	University of Georgia, algebraic geometry seminar	September 2013
	Georgia State University, commutative algebra seminar	September 2013
	Johns Hopkins University, algebraic geometry/number theory seminar	October 2013
	Georgia Algebraic Geometry Symposium, a regional meeting	October 2013
	Birational Geometry and Singularities in Positive Characteristic, a conference at the University of Tokyo	November 2013
	Mini-workshop on Cremona groups – University of Utah	November 2013
	University of Illinois at Chicago algebraic geometry seminar	November 2013
	Joint mathematical meetings in Baltimore, special session on algebraic geometry	January 2014
	University of California at Irvine, colloquium	January 2014
	University of California at Davis, colloquium	January 2014
	University of Utah, colloquium	February 2014
	University of California at Berkeley, algebraic geometry seminar	February 2014
	Birational Geometry and Foliations workshop – Hausdorff Research Institute for Mathematics	February 2014
	Queens University, algebraic geometry seminar	March 2014
	University of South Carolina, colloquium	March 2014
	Texas Algebraic Geometry Symposium (TAGS)	March 2014
	George Mason University, colloquium	April 2014

Talks continued on the next page.

SELECTED TALKS CONTINUED	George Mason University, combinatorics and algebra seminar	April 2014
	Penn State University, algebra and number theory seminar	April 2014
	Introduction to Cartier modules (3 lectures), Special Month on “Birational Geometry and Singularities in Zero and Positive Characteristic” held at the University of Michigan.	June 2014
	Research presentation, Special Month on “Birational Geometry and Singularities in Zero and Positive Characteristic” held at the University of Michigan.	June 2014
	University of Utah, algebraic geometry seminar	September 2014
	Route 81 conference – Cornell University	September 2014
	Western algebraic geometry seminar (WAGS conference) – University of Idaho	October 2014
	Georgia algebraic geometry symposium (GAGS conference) – University of Georgia	October 2014
	Georgia Tech, algebra seminar	October 2014
	University of Michigan, algebraic geometry seminar	January 2015
	Georgetown University AMS Meeting, Special Session on Closure Operations in Commutative Algebra	March 2015
	Conference on Frobenius Operators and Cartier Algebras – Georgia State University	March 2015
	AMS summer institute in Algebraic Geometry – University of Utah	July 2015
	Mini-course at the Multiplier ideals, Test ideals and Bernstein-Sato polynomials, conference – Universitat Politècnica de Catalunya, Barcelona	September 2015
	Joint mathematical meetings in Seattle, special session on commutative algebra,	January 2016
	University of Illinois at Chicago, algebraic geometry seminar	April 2016
	Differential forms in algebraic geometry (conference) – University of Freiburg, Germany	September 2016
	KUMUNU – regional commutative algebra conference	October 2016
	Nihon University Singularities Seminar	November 2016
	University of Tokyo Algebraic Geometry Seminar	November 2016
	SLAM (Southwest Local Algebra Meeting) – University of New Mexico	March 2017
	AGNES (Algebraic Geometry Northeastern Series) – Stony Brook University	April 2017

Talks continued on the next page.

SELECTED TALKS CONTINUED	London Geometry and Topology Seminar	May 2017
	Higher Dimensional Algebraic Geometry 2017, Taipei, Taiwan	June 2017
	The Prospects for Commutative Algebra, Osaka, Japan	July 2017
	University of North Texas AMS Meeting, Special Session on Commutative Algebra, Denton, Texas	September 2017
	EPFL Lausanne, Algebraic Geometry Seminar	September 2017
	Algebraic Geometry: Birational Classification, Derived Categories, and Moduli Spaces, Oberwolfach, Germany	September 2017
	University of Arizona, Algebraic Geometry Seminar	November 2017
	Mexican National Congress of Algebraic Geometry, Plenary Lecture	February 2018
	Math Science Research Institute, Hot Topics Workshop: The Homological Conjectures, MSRI, Berkeley	March 2018
	Lecture Series, Tianyuan Advanced Seminar on Moduli Spaces in Algebraic Geometry	April 2018
	Lecture series, Derived Algebraic Geometry and Birational Geometry and Moduli Spaces – MSRI, Berkeley, CA	January 2019
	University of Washington, Algebraic Geometry Seminar	May 2018
	University of Illinois at Chicago, Commutative Algebra Seminar	October 2018
	Purdue University, Commutative Algebra Seminar	October 2018
	University of Arkansas AMS Meeting, Special Session on Advances in Birational Geometry, Fayetteville Arkansas	November 2018
	University of Michigan, Commutative Algebra Seminar	November 2018
	University of Michigan, Algebraic Geometry Seminar	November 2018
	Lecture Series, Introductory Workshop: Derived Algebraic Geometry and Birational Geometry and Moduli Spaces, Math Science Research Institute (MSRI), Berkeley	January 2019
	University of Hawaii AMS Meeting, Special Session on Commutative Algebra and its Environs, Honolulu Hawaii.	March 2019
	Workshop on Algebraic Geometry, Fudan University, Shanghai, China.	July 2019
	University of Michigan, Algebraic Geometry Seminar	October 2019
	Johns Hopkins University, Algebraic Geometry Seminar	October 2019
	Joint mathematical meetings in Denver, special session on commutative algebra	January 2020
	Virtual joint mathematical meetings, special session on commutative algebra in positive characteristic	January 2021

Talks continued on the next page.

SELECTED TALKS CONTINUED	Geometry Seminar, Roma Tres (Virtual)	April 2021
	ZAG (Zoom Algebraic Geometry) Seminar (Virtual)	April 2021
	Tata Institute for Fundamental Research, Colloquium (Virtual)	August 2021
	University of Washington, Algebra and Algebraic Geometry Seminar	October 2021
	University of Illinois at Chicago, Algebraic Geometry Seminar	October 2021
	Michigan State University, Algebraic Geometry Seminar	November 2021
	Georgia Algebraic Geometry Symposium, Emory University	April 2022
	Algebraic geometry and singularities workshop and conference, University of Washington	June 2022
	CIMAT, Guanajuato Mexico, colloquium and two seminars	August 2022
	MMP and Moduli Conference, Simons Center, NY	October 2022
	Algebra Seminar, George Mason University	May 2023
	Algebraic geometry and cohomology in mixed characteristic Conference, Northwestern University	May 2023
	Lecture series, Special Month on Singularities, University of Michigan	May 2023
	Joint mathematical meetings in San Fransisco, special session on Recent Developments in Commutative Algebra	January 2024
	Taters (Topics in Algebra, Topology etc) Seminar at Boise State (Virtual)	January 2024
	University of Washington, Algebra and Algebraic Geometry Seminar	March 2024
	Workshop on p-adic Arithmetic Geometry (Spring), IAS School of Mathematics, Princeton	March 2024
	Algebra and Number Theory Day, University of Maryland	April 2024
	CAAGTUS (Commutative Algebra and Algebraic Geometry in TUcSon), University of Arizona	May 2024
	SLMath Programmatic Seminar, SLMath, Berkeley, CA	May 2024
	Commutative Algebra and Singularity Theory Conference (in Honor of the 80th Birthday of Professor Kei-ichi Watanabe), Osaka Japan	September 2024
	Methods in Mixed Charateristic Geometry, Fall School, Mainz, Germany	October 2024
	Higher Du Bois and higher rational singularities, American Institute of Mathematics	October 2024

Talks continued on the next page.

SELECTED TALKS CONTINUED	Moduli of Varieties conference, University of Utah	November 2024
	University of Arkansas, Algebra Seminar	February 2025
	CA+, Conference at Iowa State University	March 2025
	Perfection Conference, Simons Center, NY	March 2025
	Rayfest (conference in honor of Ray Heitmann's retirement), Conference at University of Nebraska	April 2025
	P-adic and Characteristic p Methods in Algebraic Geometry, EPFL, Switzerland	June 2025
	Plenary talk, Summer Research Institute in Algebraic Geometry, Colorado State University	July 2025
	Joint mathematical meetings in Washington D.C., special session on commutative algebra	January 2026
	University of Missouri, Algebra Seminar	February 2026
	Basque Center For Applied Mathematics (BCAM), Algebra Seminar	March 2026
UPCOMING TALKS	CIRM, Luminy, The Ubiquity of Commutative Algebra	March 2026
	PASCA 2 conference, Lecture Series, CIMAT, Guanajuato Mexico	May 2026